ECE 329 Fields and Waves I Summer 15

Kudeki

Exam 2

Wed, July 12, 2017 — 11 AM - 11:55

Name:	Solution
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a closed book exam and calculators are not allowed. You are allowed two formula sheets of 8.5 by 11 inch dimensions — both sides of the sheets may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

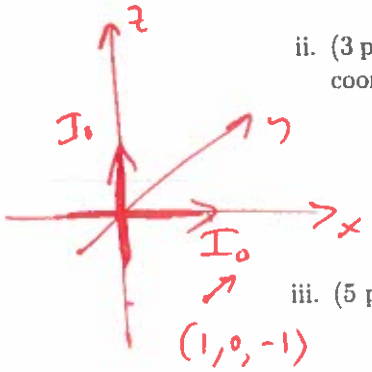
Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. **Magnetic fields of line currents and solenoids:**

a) Consider the DC current density function $\mathbf{J}(x,y,z) = I_o[\delta(x)\delta(y)\,\hat{z} + \delta(y)\delta(z)\,\hat{x}]$ A/m^2 where coordinates x, y, and z are measured in meter units.

i. (2 pts) What are the units of parameter I_o in the MKSA system? Justify your answer.

$I_0 [=] A$ so that $\vec{J} [=] \frac{A}{m^2}$ given that $\delta(x)\,\delta(y) [=] \frac{1}{m^2}$



ii. (3 pts) Sketch a diagram depicting the current system described by $\mathbf{J}(x,y,z)$ in the Cartesian coordinate system indicating the locations and directions of current flows.

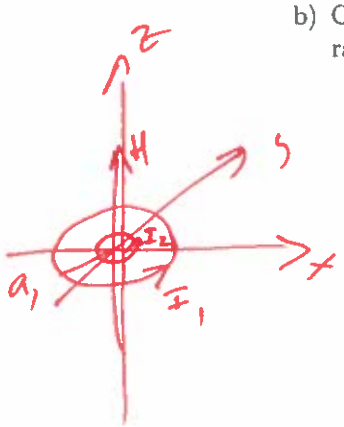
$$\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{\phi}$$

iii. (5 pts) If $\mathbf{H}(1,0,-1) = 4\hat{y}$ A/m is caused by $\mathbf{J}(x,y,z)$, determine I_o? Show your work.

$$H(1,0,-1) = 4\hat{y}\,\frac{A}{m} = 2\hat{y}\,\frac{I_0}{2\pi \cdot 1} \Rightarrow I_0 = 4\pi\ A$$

superposition.

b) Consider a long solenoid around and parallel to z-axis having $N = 1000$ loops/meter and a radius of $a_1 = 1$ m.



i. (5 pts) If the solenoid current is $I_1 = 1$ mA and this generates an internal magnetic field intensity $\mathbf{H} = H_o\hat{z}$ determine H_o (with the appropriate unit)

$$N I_1 = (1000)(10^{-3})\,\frac{A}{m} = 1\,\frac{A}{m} = J_s$$

$H_0 = 1\,\frac{A}{m}$ assuming I_1 as shown.

ii. (5 pts) Another solenoid with $N = 1000$ loops/meter and $a_2 = 0.5$ m is inserted within the first solenoid and then its current I_2 is adjusted so that $\mathbf{H} = 0$ for $r < a_2$. Describe I_2 needed to accomplish this in comparison to the magnitude and direction of I_1.

$I_2 = -I_1 = -1\text{mA}$. I_2 has same magnitude but opposite direction as I_1

iii. (5 pts) If I_1 slowly grows in time so that H_o grows in proportion, how will I_2 respond to this change according to Faraday's law and Lenz's rule briefly describe.

$I_1 \uparrow$, $\Psi_1 \uparrow$ $\frac{d\Psi_1}{dt} > 0$, $\mathcal{E} = -\frac{d\Psi_1}{dt} < 0$

I_2, in opposite direction will also grow to oppose the growth of flux within.

2. **Two parts of this question are independent:**

a) In a homogeneous conductor where $\mathbf{J} = \sigma\mathbf{E}$, Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_o}$ and the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$ can be used together to derive a differential equation

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_o}\rho = 0$$

for the charge density ρ.

i. (6 pts) Find the solution of the differential equation above for $t > 0$ if at $t = 0$ the charge density is $\rho(x, y, z, 0) = \cos(3z)$ C/m^3 over all space.

$\rho(x,y,z,t) = \cos(3z)\, e^{-t/\tau}$, $\tau = \sigma/\epsilon_o$

ii. (6 pts)Given the solution of part (i) for $\rho(x, y, z, t)$ and Gauss's Law $\nabla \cdot \mathbf{E} = \rho/\epsilon_o$ find an electric field solution $\mathbf{E}(x, y, z, t)$ having only a z-component. **Hint:** First determine $\partial E_z/\partial z$ and then find its anti-derivative.

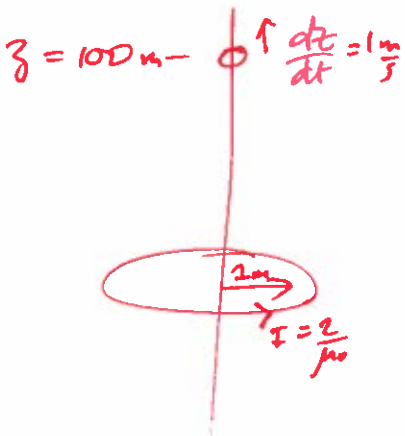
$\nabla \cdot E = \frac{\partial E_z}{\partial z} = \frac{\rho}{\epsilon_o} = \frac{\cos(3z)}{\epsilon_o} e^{-t/\tau}$

$\Rightarrow E_z = \frac{\sin(3z)}{3\epsilon_o} e^{-t/\tau}$

b) (13 pts) Given a loop of wire with radius a_1 on the xy-plane and current I in counterclockwise direction when viewed from the top, the magnetic field evaluated on the z-axis is given as

$$\mathbf{B} = \hat{z}\frac{\mu_o I a_1^2}{2z^3}$$

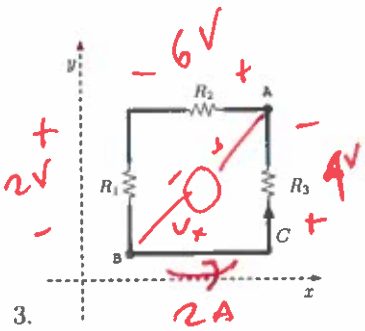
for $z \gg a_1$. Let $a_1 = 1$ m, $I = 2/\mu_o$, and use this magnetic field vector to estimate the induced emf (in volts) of a small loop with radius a_2=1 cm, encircling the z-axis, and moving upwards with velocity $dz/dt = 1$ m/s in the vicinity of $z = 100$ m. Simplify your result as much as you can.



$\Psi = (\pi a_2^2)\left(\frac{\mu_o I a_1^2}{2z^3}\right) = \pi (10^{-2})^2 \frac{(1)^2}{z^3}$

$\mathcal{E} = -\frac{\partial \Psi}{\partial t} = -\pi 10^{-4}(-3)\frac{1}{z^4}\frac{dz}{dt}$

$= 3\pi \frac{10^{-4}}{(10^2)^4} 1 = 3\pi \times 10^{-12}$ V



3. Consider the circuit depicted on the left. Resistors $R_1 = 1\,\Omega$, $R_2 = 3\,\Omega$, and $R_3 = 2\,\Omega$ constitute a square loop of 1 m^2 physical area in a region where there is a uniform magnetic field $\mathbf{B} = B_o\hat{z}$. It is known that $dB_o/dt = -12$ T/s. Answer the following questions:

a) (5 pts) Compute the induced emf $\mathcal{E}$ in the circuit in the circulation direction C indicated on the diagram.

$\Psi_o = (1\,m^2)B_o \Rightarrow \mathcal{E} = -\frac{d\Psi}{dt} = -\frac{dB_o}{dt} \leftarrow -12\frac{T}{s} = +12\,V$

b) (5 pts) Compute the loop current I induced in the circuit in the circulation direction C indicated on the diagram.

$R_{tot} = 6\Omega \qquad I = \frac{12V}{6\Omega} = 2A$

c) (10 pts) If a voltmeter is connected *diagonally* between points A and B with the (+) lead contacting point A (and in effect splitting the square loop into a pair of equal area triangles) what will be the voltmeter reading?

$V_x + 4 = \frac{12}{2} = 6 \Rightarrow V_x = 2V$

d) (5 pts) Describe how you would re-connect the voltmeter between points A and B to directly read the voltage drop of R_3 on the right.

